

Black Holes

Black holes are regions in space where gravity is so strong that nothing—not even light—can escape. They form when massive stars collapse under their own gravity at the end of their life cycles. **Key**

Features:

- **Event Horizon:** The boundary beyond which nothing can return.
- **Singularity:** A point of infinite density at the center of a black hole.
- **Accretion Disk:** Hot, glowing matter spiraling into the black hole.

Types of Black Holes:

- *Stellar Black Holes:* Formed from collapsed stars.
- *Supermassive Black Holes:* Found at the centers of galaxies.
- *Intermediate Black Holes:* Mid-sized, not well understood.

Interesting Facts:

- Time slows down near a black hole due to extreme gravity.
- They can merge, releasing enormous gravitational waves.
- They do not “suck” objects; instead, they attract them like any massive body.

Black holes remain one of the greatest mysteries of the universe, offering insights into gravity, spacetime, and quantum physics.